

FIFA World Cup 2026

Simulation Forecasts Versus Actual Results

Prepared 8 October 2026 | Four models | 10,000 runs per model

Executive Overview

This report evaluates frozen June simulation outputs against the tournament outcomes recorded by FIFA. It compares title probabilities, advancement forecasts, model scores and expected scoring leaders. No model was retrained for this evaluation.

Spain won the final against Argentina, 1–0 after extra time. England finished third and France fourth. Kylian Mbappe won the Golden Boot with 10 goals.

Principal Findings

- The three primary models assigned Spain title probabilities of 10.26%, 18.18% and 17.11%. Spain ranked second in the baseline and fourth in both v2 models.
- None of the primary models ranked the eventual champion first: the baseline favoured France, while both v2 models favoured Brazil.
- Heuristic v2 had the lowest mean advancement Brier score in this tournament: 0.0840, compared with 0.0872 for the baseline and 0.0867 for Random Forest v2.
- Germany and Brazil exited earlier than their v2 forecasts suggested. Norway and Morocco exceeded their predicted advancement chances.

Scope of the Evidence

These are descriptive results from one tournament. The saved v2 outputs used FIFA-rank-derived rating priors and synthetic training data. The comparison is not an independently audited pre-tournament backtest, and it does not establish general predictive superiority.

Reading Guide

Section 1 explains the evaluation; Section 2 compares championship probabilities; Section 3 evaluates advancement; Sections 4 and 5 examine model discrepancies and player scoring. Section 6 documents limitations, future work and sources.

1 Evaluation Design

The comparison joins frozen forecast percentages with binary indicators of actual advancement for each of the 48 teams. The analysis verifies that the comparison CSVs match the original JSON outputs and that the observed stage memberships are nested correctly.

How the comparison was made

Saved forecasts are evaluated without retraining.



Outcome checks: 32 -> 16 -> 8 -> 4 -> 2 -> 1; CSV forecasts checked against original JSON.

Figure 1. Workflow for evaluating saved forecasts against sourced tournament outcomes.

1.1 Probability Scores

For team t at stage s , let $p_{t,s}$ be the forecast probability and $y_{t,s} \in \{0, 1\}$ indicate whether that team reached the stage.

$$BS_s = \frac{1}{48} \sum_{t=1}^{48} (p_{t,s} - y_{t,s})^2, \quad (1)$$

$$\text{Skill}_s = 1 - \frac{BS_s}{q_s(1 - q_s)}, \quad q_s = \frac{k_s}{48}, \quad (2)$$

$$LL_{\text{champion}} = -\ln p_{\text{Spain, champion}}. \quad (3)$$

Lower Brier score and log loss are better. Positive skill indicates improvement over a uniform stage probability. The categorical champion Brier score is the *sum* across 48 outcomes; advancement Brier scores are *means*.

The combined advancement score is the equally weighted mean over six stages: last 32, last 16, quarter-finals, semi-finals, final and champion. Third- and fourth-place outcomes are reported separately.

1.2 Top- k Overlap

For a stage admitting k teams, the analysis counts how many of the k highest forecast teams actually reached it. Equal probabilities use alphabetical ordering. This ranking measure complements probability scores but does not assess calibration.

1.3 Model Inventory

The primary comparison includes the EAFC/Poisson baseline, heuristic v2 and Random Forest v2. The earlier squad-proxy Random Forest is retained as a legacy reference and excluded from the primary aggregate comparison.

2 Championship Forecasts

Spain had substantial probability in each primary forecast, but no primary model selected Spain as its favourite. A likely contender winning is compatible with the forecast even when the highest-ranked team's prediction is incorrect.

Who each model expected to win

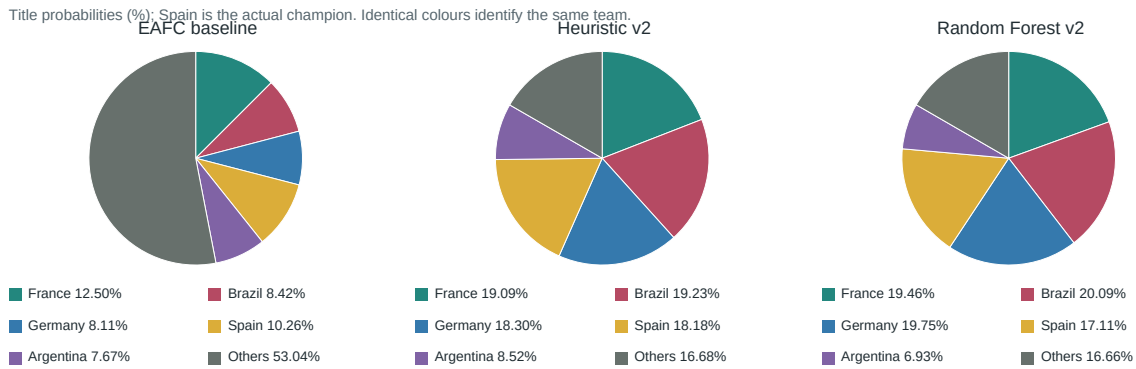


Figure 2. Title-probability distributions for the three primary models. Each pie totals 100%; Others combines all teams outside the five named contenders.

Table 1. Forecast support for the actual champion. Legacy RF is a reference model.

Model	Favourite	Spain rank	Spain (%)	Log loss	Brier sum
EAFC baseline	France	2	10.26	2.277	0.864
Heuristic v2	Brazil	4	18.18	1.705	0.790
Random Forest v2	Brazil	4	17.11	1.766	0.817
Legacy RF	France	2	14.38	1.939	0.870

Spain's forecast title probability

Spain won; probabilities describe the forecast before the observed outcome.

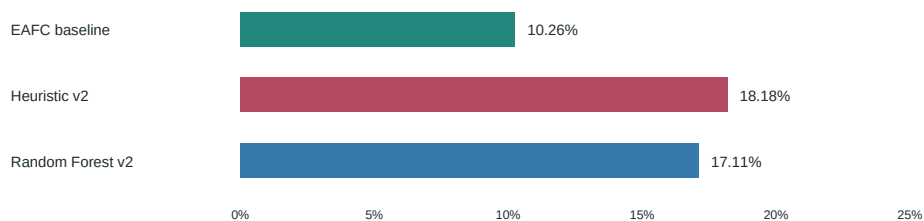


Figure 3. Spain's title probability across the primary models.

3 Advancement Accuracy

The richer heuristic model improves the combined score relative to the baseline, but the improvement is not uniform across stages. The baseline has lower Brier error at the quarter-final, semi-final and final stages.

Forecast error across tournament stages

Mean Brier score across 48 teams; lower is better. Six stages receive equal weight.

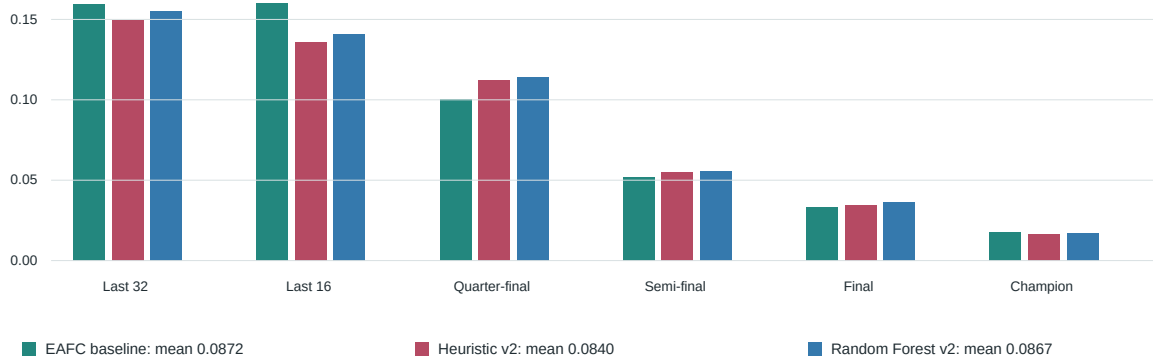


Figure 4. Stage-wise mean Brier scores. Compare models within a stage because the stage base rates differ.

Table 2. Advancement error and top-ranked qualifier overlap across the three primary models.

Stage	Baseline	Heuristic	RF	Top- <i>k</i> hits (B / H / RF)
Last 32	0.1594	0.1497	0.1551	27/32 / 27/32 / 27/32
Last 16	0.1599	0.1357	0.1411	11/16 / 11/16 / 11/16
Quarter-final	0.1004	0.1124	0.1143	4/8 / 4/8 / 4/8
Semi-final	0.0521	0.0552	0.0559	2/4 / 2/4 / 2/4
Final	0.0332	0.0345	0.0366	1/2 / 0/2 / 0/2
Champion	0.0180	0.0165	0.0170	0/1 / 0/1 / 0/1
Third place	0.0190	0.0187	0.0174	0/1 / 0/1 / 0/1
Fourth place	0.0197	0.0211	0.0211	0/1 / 0/1 / 0/1

Table 3. Equal-weight combined advancement scores; lower is better.

Primary model	Mean six-stage Brier
EAFC baseline	0.0872
Heuristic v2	0.0840
Random Forest v2	0.0867

4 Where Forecasts and Outcomes Diverged

Forecasts versus actual advancement

Random Forest v2; outlined cells indicate teams that actually reached the stage.

	Last 32	Last 16	Quarter-final	Semi-final	Final	Champion
Spain	99.8%	88.5%	53.0%	39.2%	26.6%	17.1%
Argentina	99.9%	83.6%	57.5%	37.0%	14.2%	6.9%
England	99.9%	92.7%	75.4%	34.5%	13.5%	6.8%
France	99.8%	95.5%	52.1%	40.0%	29.4%	19.5%
Brazil	99.8%	86.8%	78.0%	68.8%	42.1%	20.1%
Germany	99.8%	93.1%	82.2%	72.4%	42.9%	19.8%
Norway	89.9%	48.0%	11.4%	3.5%	0.8%	0.1%
Morocco	91.0%	27.6%	12.6%	4.2%	0.8%	0.1%
Belgium	99.9%	90.2%	39.9%	10.2%	3.7%	1.1%
Switzerland	98.2%	69.2%	20.9%	8.2%	1.8%	0.2%

Darker fill = higher forecast probability. Outline = actual advancement. This is a selected-team view.

Figure 5. Random Forest v2 probabilities for selected teams. Outlined cells indicate actual advancement; darker fills indicate higher forecast probability.

Germany exited in the last 32 and Brazil in the last 16 despite high v2 advancement probabilities. Norway, Morocco, Belgium and Switzerland reached the quarter-finals. The selected-team heatmap illustrates both overestimation of highly rated contenders and underestimation of unexpected progress.

These discrepancies suggest reviewing upset frequency, bracket effects and player-data coverage. The observed outcomes alone cannot identify which mechanism caused a forecast error.

Table 4. Eight largest absolute discrepancies among primary-model quarter-final, semi-final and final forecasts.

Model	Team	Stage	Forecast (%)	Reached?
Heuristic v2	Norway	Quarter-final	10.38	Yes
Random Forest v2	Norway	Quarter-final	11.41	Yes
Random Forest v2	Morocco	Quarter-final	12.61	Yes
EAFC baseline	Argentina	Final	13.86	Yes
Random Forest v2	Argentina	Final	14.16	Yes
Heuristic v2	Morocco	Quarter-final	15.66	Yes
Heuristic v2	Argentina	Final	17.25	Yes
EAFC baseline	Spain	Final	17.77	Yes

5 Player Scoring and Interpretation

Kylian Mbappe won the Golden Boot with 10 goals. Both primary v2 models ranked him first by expected aggregate goals; the baseline ranked him second.

Table 5. Expected scoring rank and mean goals versus one observed tournament.

Model	Mbappe rank	Mean goals	Actual goals
EAFC baseline	2	1.446	10
Heuristic v2	1	2.946	10
Random Forest v2	1	2.904	10
Legacy RF	1	3.467	10

Expected goals over many simulated tournaments and observed goals in a single tournament are different quantities. A mean of approximately three goals does not predict that the player must score three. Similarly, a ranking by expected goals is not the probability of winning the Golden Boot. This comparison is descriptive; award forecasting would require a separate probability calculation.

6 Provenance, Limitations and Next Steps

6.1 Forecast Provenance

Each model ran 10,000 tournaments. The two primary v2 artifacts used FIFA-rank-derived Elo-scale priors, rather than the raw World Football Elo data imported later. Random Forest training used synthetic profile calibration; historical-match predictive accuracy was not established.

Seeds that resemble dates are random-number seeds, not creation timestamps. The artifacts were saved around the start of the tournament and the v2 files were last modified on 11 June. Local timestamps do not prove publication before kickoff. This report therefore evaluates saved forecasts retrospectively.

6.2 Statistical and Data Limits

Advancement indicators are dependent; six stages across 48 teams are not 288 independent observations. One tournament cannot establish calibration or model superiority. Wilson intervals in prior reports measure Monte Carlo sampling error conditional on model assumptions, not uncertainty about those assumptions.

Saved outputs do not supply probabilities for every actual match or every player. Match-level log loss and full player-goal errors are outside scope. Group-stage exits are inferred from absence in the verified last-32 field. Rankings among teams eliminated at the same stage are not inferred.

6.3 Recommended Research Work

1. Backtest several historical tournaments using information available at each match date.
2. Calibrate goal, draw and upset rates on real international results.
3. Review bracket routing and concentration of probability on stronger teams.
4. Improve low-confidence player coverage and model Golden Boot probabilities separately.
5. Publish dated forecast snapshots before future tournaments.

6.4 Reproducibility

Run `python compare_simulation_with_actuals.py` from the repository root. The analysis regenerates CSV score tables, Markdown and HTML reports, charts and this LaTeX source. Compile the source from the report folder using `pdflatex comparison_report.tex` twice for cross-references.

Supporting data: `actual_outcomes.json`, `model_scores.csv`,
`team_stage_comparison.csv`, `champion_comparison.csv`,
`scorer_comparison.csv`, `prediction_sha256.json`.

The hashes identify the frozen inputs. Figure PDFs are supplied in the `figures/` folder.

6.5 Sources

Tournament outcomes were verified against the following FIFA pages on 8 October 2026. Source links and team-name mappings are retained in `actual_outcomes.json`.

1. Knockout fixtures and results.
<https://www.fifa.com/en/articles/knockout-stage-match-schedule-bracket?searchOverlay=1>
2. Final tournament standings.
<https://www.fifa.com/en/articles/final-tournament-standings>
3. France campaign and Golden Boot result.
<https://www.fifa.com/en/articles/lessons-from-frances-2026-world-cup-campaign>
4. Final and champion confirmation.
<https://inside.fifa.com/organisation/news/new-york-jersey-stadium-spain-world-champions-mbappe-haaland>